LECTURE 7

Text Wrangling and Regex

Using string methods and regular expressions (regex) to work with textual data

Presented by Dr. Devan Becker, Data200

Adapted from Data 100, Summer 2023 @ UC Berkeley by Bella Crouch and Dominic Liu

Content credit: Lisa Yan, Josh Hug

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Announcements

- A1 is marked!
- Project is out!

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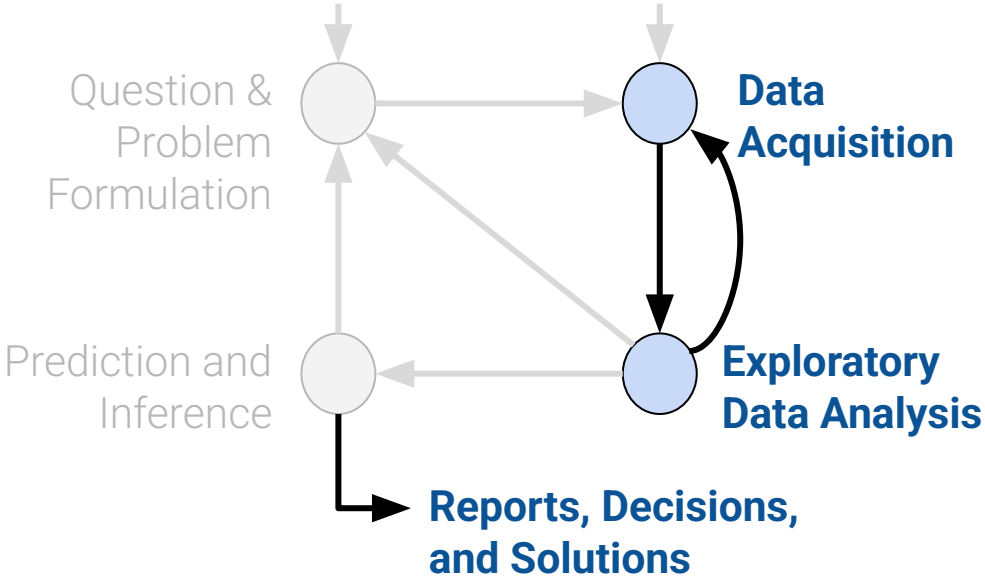
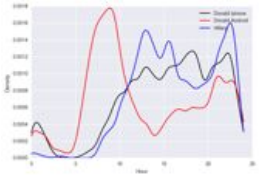
This week





Data Wrangling
Intro to EDA

→

Working with Text Data
Regular Expressions

→

Visualization
Code for plotting data

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Goals for this Lecture

Deal with a major challenge of EDA: cleaning text

- Operate on text data using `str` methods
- Apply regex to identify patterns in strings

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Agenda

- Why work with text?
- `pandas str` methods
- Why regex?
- Regex basics
- Regex functions

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Why Work With Text?

- **Why work with text?**
- `pandas str` methods
- Why regex?
- Regex basics
- Regex functions

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Why work with text? Two common goals

1. **Canonicalization**: Convert data that has more than one possible presentation into a standard form.

Ex Join tables with mismatched labels

	County	State
0	De Witt County	IL
1	Lac qui Parle County	MN
2	Lewis and Clark County	MT
3	St John the Baptist Parish	LA

	County	Population
0	DeWitt	16798
1	Lac Qui Parle	8067
2	Lewis & Clark	55716
3	St. John the Baptist	43044

	County	State	Population
0	dewitt	IL	16798
1	lacquiparle	MN	8067
2	lewisandclark	MT	55716
3	stjohnthebaptist	LS	43044

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Why work with text? Two common goals

1. **Canonicalization**: Convert data that has more than one possible presentation into a standard form.

Ex Join tables with mismatched labels

	County	State
0	De Witt County	IL
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	County	Population
0	DeWitt	16798
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join?

	County	Population	State
0	dewitt	16798	IL
1	lacquiparle	8067	MN
2	lewisandclark	55716	MT
3	stjohnthebaptist	43044	LS

2. **Extract** information into a new feature.

Ex Extract dates and times from log files

```
169.237.46.168 - -
[26/Jan/2014:10:47:58 -0800] "GET
/stat141/Winter04/ HTTP/1.1" 200 2585
"http://anson.ucdavis.edu/courses/"
```

```
day, month, year = "26", "Jan", "2014"
hour, minute, seconds = "10", "47", "58"
```


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`pandas str` methods

- Why work with text?
- **`pandas str` methods**
- Why regex?
- Regex basics
- Regex functions

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From String to `str`

In “base” Python, we have various string operations to work with text data.

Recall:

transformation	`s.lower()` `s.upper()`
split	`s.split(…)`
membership	`'ab' in s`
replacement/ deletion	`s.replace(…)`
substring	`s[1:4]`
length	`len(s)`

Problem: Python assumes we are working with one string at a time
Need to loop over each entry – slow in large datasets!

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str methods

Fortunately, pandas offers a method of **vectorizing** text operations: the `.str` operator

```
Series.str.string_operation()
```

Apply the function `string_operation` to *every* string contained in the `Series`

```
populations["County"].str.lower()
```

```
0                   dewitt
1            lac qui parle
2            lewis & clark
3    st. john the baptist
Name: County, dtype: object
```

```
populations["County"].str.replace('&', 'and')
```

```
0                   DeWitt
1            Lac Qui Parle
2          Lewis and Clark
3    St. John the Baptist
Name: County, dtype: object
```


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`.str` methods

Most base Python string operations have a `pandas str` equivalent

Operation	Python (single string)	pandas (Series of strings)
transformation	`s.lower()` `s.upper()`	`ser.str.lower()` `ser.str.upper()`
replacement/ deletion	`s.replace(…)`	`ser.str.replace(…)`
split	`s.split(…)`	`ser.str.split(…)`
substring	`s[1:4]`	`ser.str[1:4]`
membership	`'ab' in s`	`ser.str.contains(…)`
length	`len(s)`	`ser.str.len()`

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Demo Slides

[UC Berkeley Demo]

Canonicalization

	County	State
0	De Witt County	IL
1	Lac qui Parle County	MN
2	Lewis and Clark County	MT
3	St John the Baptist Parish	LA

	County	Population
0	DeWitt	16798
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	County	State	Population
0	dewitt	IL	16798
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3	stjohnthebaptist	LS	43044

```
def canonicalize_county(county_series):
 return (county_series
                .str.lower()                # lowercase
                .str.replace(' ', '')       # remove space
                .str.replace('&', 'and')    # replace &
                .str.replace('.', '')       # remove dot
                .str.replace('county', '')
                .str.replace('parish', '')
          )
```

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Why regex?

- Why work with text?
- `pandas str` methods
- **Why regex?**
- Regex basics
- Regex functions

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Flexibility matters!

This seems great…but there's still a lot of time left in lecture 🧐

We made a big assumption in the previous example: knowing for certain what changes needed to be made to the text. "Eyeballing" the steps needed for canonicalization.

Consider our data extraction task from before – pulling out dates from log data:

```
169.237.46.168 - -
[26/Jan/2014:10:47:58 -0800] "GET
/stat141/Winter04/ HTTP/1.1" 200 2585
"http://anson.ucdavis.edu/courses/"
```

```
193.205.203.3 - -
[2/Feb/2005:17:23:6 -0800] "GET
/stat141/Notes/dim.html HTTP/1.0" 404 302
"http://eeyore.ucdavis.edu/stat141/Notes/
session.html"
```


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Flexibility matters!

```
169.237.46.168 - -
[26/Jan/2014:10:47:58 -0800] "GET
/stat141/Winter04/ HTTP/1.1" 200 2585
"http://anson.ucdavis.edu/courses/"
```

```
193.205.203.3 - -
[2/Feb/2005:17:23:6 -0800] "GET
/stat141/Notes/dim.html HTTP/1.0" 404 302
"http://eeyore.ucdavis.edu/stat141/Notes/
session.html"
```

Formatting varies:

- Different # of characters before the date
- Different format for the day of the month
- Different # of characters after the date

We don't always know the exact format of our data in advance.

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Demo Slides

Extracting Date Information

```
169.237.46.168 - -
[26/Jan/2014:10:47:58 -0800] "GET
/stat141/Winter04/ HTTP/1.1" 200 2585
"http://anson.ucdavis.edu/courses/"
```

```
day, month, year = "26", "Jan", "2014"
hour, minute, seconds = "10", "47", "58"
```

One possible solution:

```
pertinent = line.split("[")[1].split(']')[0]
day, month, rest          = pertinent.split('/')
year, hour, minute, rest = rest.split(':')
seconds, time_zone        = rest.split(' ')
```


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String extraction: an alternate approach

While we can hack together code that uses **`replace/split`**…

```
pertinent = line.split("[")[1].split(']')[0]
day, month, rest = pertinent.split('/')
year, hour, minute, rest = rest.split(':')
seconds, time_zone = rest.split(' ')
```

An alternate approach is to use a **regular expression**:

- Implementation provided in the Python **`re`** library and the pandas **`str`** accessor.
- Next, we'll spend some time working up to expressions like this one:

```
import re
pattern = r'\[(\d+)\/(\w+)\/(\d+):(\d+):(\d+):(\d+) (.+)\]'
day, month, year, hour, minute, second, time_zone = re.findall(pattern, line)[0]
```

Related: How would you extract all the **moon**-like patterns in this string?

```
"moon moo moooooon mon moooon"
```

Seem impossible?

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Regex basics

- Why work with text?
- `pandas str` methods
- Why regex?
- **Regex basics**
- Regex functions

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A regular expression describes a set of strings

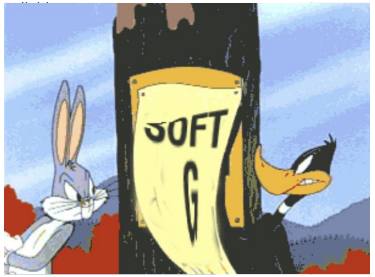
A regular expression (“regex”) is a sequence of characters that specifies a search pattern.

Example: `[0-9]{3}-[0-9]{2}-[0-9]{4}`

3 of any digit, then a dash,
then 2 of any digit, then a dash,
then 4 of any digit.

The language of Social Security Numbers is described by this regular expression.

Formal language, described implicitly



“Regex” pronunciation?
Check out English Stackexchange discussion

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Goals for regex

The goal of today is NOT to memorize regex!

Instead:

1. Understand what regex is capable of.
2. Parse and create regex, **with a reference table**.

high-level

3. Use vocabulary (metacharacter, escape character, groups, etc.) to **describe regex** metacharacters.
4. **Differentiate** between `( )`, `[ ]`, `{ }`
5. Design your own **character classes** with `\d`, `\w`, `\s`, `[ …- … ]`, `^`, etc.
6. Use Python and `pandas` regex methods.

details; hone with practice

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Regex basics

There are four basic operations in regex.

Concatenation – "look for consecutive characters"

`AABAAB` matches `AABAAB`

`|` – "or"

`AA|BAAB` matches `AA` *or* `BAAB`

`*` – "zero or more"

`AB*A` matches `AA, ABA, ABBA, ...`

`()` – "consider a group"

`(AB)*A` matches `A, ABA, ABABA, ...`
`A(A|B)AAB` matches `AAAAB` *or* `ABAAB`

`*`, `(` `)`, and `|` are called **metacharacters** – they represent an operation, rather than a literal text character

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Resources for practicing regex

There are many nice resources out there to experiment with regular expressions (e.g. regex101.com, regexone.com, Sublime Text).

I recommend trying out [regex101.com,](https://regex101.com) which provides a visually appealing and easy to use platform for experimenting with regular expressions.

- **Important:** choose the Python “flavor” in the left sidebar

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Summary so far

Operation	Order	Example	Matches	Doesn't match
concatenation (consecutive chars)	3	AABAAB	AABAAB	every other string
or, |	4	AA|BAAB	AA BAAB	every other string
***** (zero or more)	2	AB*A	AA ABBBBBBA	AB ABABA
group (parenthesis)	1	A(A|B)AAB	AAAAB ABAAB	every other string
		(AB)*A	A ABABABABA	AA ABBA

The regex order of operations. Grouping is evaluated first.

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Regex expanded

Five more regex operations.

`.` – "look for *any* character"

`.U.U.U.` matches `CUMULUS, JUGULUM`

`+` – "one or more"

`AB+` matches `AB, ABB, ABB, ...`

`{x}` – "repeat exactly x times"

`AB{2}` matches `ABB`

`{x, y}` – "repeat between x and y times"

`AB{0,2}` matches `A, AB, ABB`

`[]` – "define a character class"

`[A-Za-z]` matches `A, a, B, b...`

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Character classes

A character class describes a set of characters belonging to the class.

[A-Z] – any uppercase letter between A and Z

[0-9] – any digit between 0 and 9

[A-Za-z0-9] – any letter, any digit

Regex built-in classes:

\w is equivalent to [A-Za-z0-9]

\d is equivalent to [0-9]

\s matches whitespace

Use **^** to negate a class = match any character *other* than what follows

[^A-Z] – anything that is *not* an uppercase letter between A and Z

Equivalently, the capital versions of the regex built-in classes are negations: `\W`, `\D`, and `\S`

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Summary so far

Operation	Example	Matches	Doesn't match
any character (except newline)	`.U.U.U.`	`CUMULUS` `JUGULUM`	`SUCCUBUS` `TUMULTUOUS`
character class	`[A-Za-z][a-z]*`	`word` `Capitalized`	`camelCase` `4illegal`
repeated exactly a times: {a}	`j[aeiou]{3}hn`	`jaoehn` `jooohn`	`jhn` `jaeiouhn`
repeated from a to b times: {a,b}	`j[ou]{1,2}hn`	`john` `juohn`	`jhn` `jooohn`
at least one	`jo+hn`	`john` `jooooohn`	`jhn` `jjohn`

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Greediness

Regex is **greedy** – it will look for the *longest possible* match in a string

```
.*[aeiou]+.*
```

"Only one word in this sentence **seems** like it should match."

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Greediness

Regex is **greedy** – it will look for the *longest possible* match in a string

```
.*[aeiou]+.*
```

In English:

- "Look for any character 0 or more times"
- then, "look for a, e, i, o, or u 1+ times"
- then, "look for any character 0 or more times"

"Only one word in this sentence **seems** like it should match."

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Greediness

Regex is **greedy** – it will look for the *longest possible* match in a string

```
.*[aeiou]+.*
```

In English:

- "Look for any character 0 or more times"
- then, "look for a, e, i, o, or u 1+ times"
- then, "look for any character 0 or more times"

"Only one word in this sentence **seems** like it should match."

This pattern greedily matches the *entire* sentence!

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Greediness

Regex is **greedy** – it will look for the *longest possible* match in a string

```
[a-z]*(aa|ee|ii|oo|uu)[a-z]*
```

In English:

- "Look for a letter a-z 0 or more times"
- then, "look for aa, ee, ii, oo, or uu"
- then, "look for a letter a-z 0 or more times"

"Only one word in this sentence **seems** like it should match."

These spaces do not obey the condition "a letter a-z"

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Regex *even more* expanded

The last set, I promise.

**** – "read the next character literally"

`a\+b` matches `a+b`

^ – "match the beginning of a string"

`^abc` does not match "`123 abc`"

$ – "match the end of a string"

`abc$` does not match "`abc 123`"

Be careful: ^ has different behavior inside/outside of character classes!

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Summary so far

Operation	Example	Matches	Doesn't match
beginning of line	`^ark`	`ark two` `ark o ark`	`dark`
end of line	`ark$`	`dark` `ark o ark`	`ark two`
escape character	`cow\.com`	`cow.com`	`cowscom`

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Interlude

Regex fun? https://alf.nu/RegexGolf

Email Address Regular Expression (probably a bad idea)

```
(?:(?:\r\n)?[ \t])*(?:(?:(?:[^()<>@,;:\\".\[\] \000-\031]+(?:(?:(?:\r\n)?[ \t])+|\Z|(?=[\["()<>@,;:\\".\[\]]))|"(?:[^"\r\n\]|\\.|(?:(?:\r\n)?[ \t]))*"(?:(?:
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()<>@,;:\\".\[\]]))|"(?:[^"\r\n\]|\\.|(?:(?:\r\n)?[ \t]))*"(?:(?:\r\n)?[ \t])*)@(?:(?:\r\n)?[ \t])*(?:[^()<>@,;:\\".\[\] \000-\031]+(?:(?:(?:\r\n)?[ \t])
+|\Z|(?=[\["()<>@,;:\\".\[\]]))|\[([^\[\]\r\\]|\\.)*\](?:(?:\r\n)?[ \t])*)(?:\.(?:(?:\r\n)?[ \t])*(?:[^()<>@,;:\\".\[\] \000-\031]+(?:(?:(?:\r\n)?[ \t])+|\Z
|(?=[\["()<>@,;:\\".\[\]]))|\[([^\[\]\r\\]|\\.)*\](?:(?:\r\n)?[ \t]))*\>(?:(?:\r\n)?[ \t])*)*;\s*)
```

source, StackOverflow discussion

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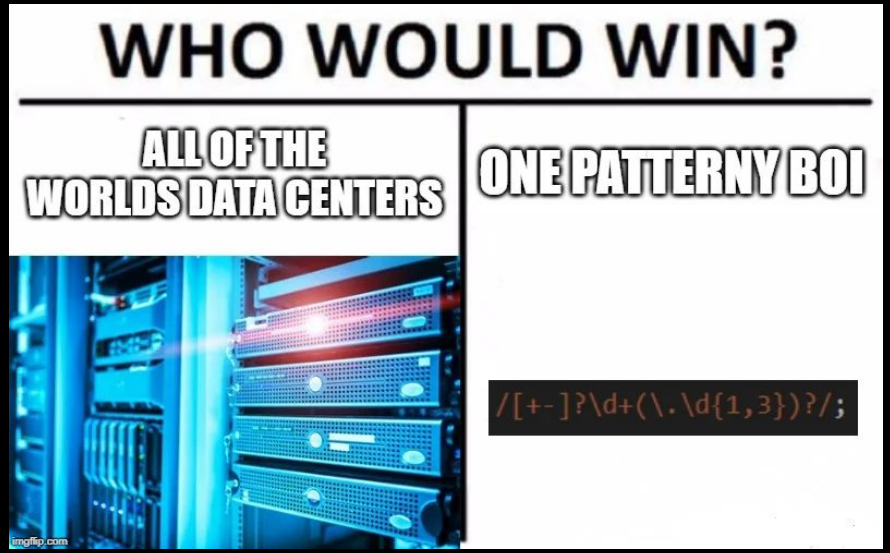

Interlude

[Cloudfare Outage]

- 30 minute failure in about 20% of the internet.
- Caused by one bad regex
  - Trying to guard against code injection
- Other problem: pushing all code everywhere all at once

BY NC SA

Regex functions

- Why work with text?
- `pandas str` methods
- Why regex?
- Regex basics
- **Regex functions**

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Before we begin: raw strings in Python

When specifying a pattern, we strongly suggest using **raw strings**.

- A raw string is created using **`r""`** or **`r''`** instead of just `""` or `''`.

```
pattern = r"[0-9]+"
```

- The exact reason is a bit tedious.
  - Rough idea: Regular expressions and Python strings both use `\` as a special character.
  - Using non-raw strings leads to uglier regular expressions.

Regular String	Raw string
`"ab*"`	`r"ab*"`
`"\\\\section"`	`r"\\section"`
`"\\w+\\s+\\1"`	`r"\w+\s+\1"`

For more information see “The Backslash Plague” under
https://docs.python.org/3/howto/regex.html#the-backslash-plague

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Extraction

`re.findall(pattern, text)` docs

Return a list of all matches to `pattern`.

```
text = "My social security number is
123-45-6789 bro, or actually maybe it's
321-45-6789.";
pattern = r"[0-9]{3}-[0-9]{2}-[0-9]{4}"
re.findall(pattern, text)
```

```
['123-45-6789', '321-45-6789']
```

A **match** is a substring that matches the provided regex.

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Extraction

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Return a list of all matches to `pattern`.

```
text = "My social security number is 123-45-6789 bro, or actually maybe it's 321-45-6789.";
pattern = r"[0-9]{3}-[0-9]{2}-[0-9]{4}"
re.findall(pattern, text)
```

```
['123-45-6789', '321-45-6789']
```

`ser.str.findall(pattern)` docs

Returns a Series of lists

```
df["SSN"].str.findall(pattern)
```

	SSN
0	987-65-4321
1	forty
2	123-45-6789 bro or 321-45-6789
3	999-99-9999

```
0                 [987-65-4321]
1                            []
2  [123-45-6789, 321-45-6789]
3                 [999-99-9999]
Name: SSN, dtype: object
```


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Extraction with capture groups

Earlier we used parentheses to specify the **order of operations**.

Parenthesis can have another meaning:

- When using certain regex functions, ( ) specifies a **capture group**.
- Extract *only* the portion of the regex pattern inside the capture group

```
text = """I will meet you at 08:30:00 pm tomorrow"""
pattern = ".*(\d\d):(\d\d):(\d\d).*"
matches = re.findall(pattern, text)
matches
```

The capture groups each capture two digits.

```
[('08', '30', '00')]
```


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Extraction with capture groups

`ser.str.extract(pattern)` docs

Returns a DataFrame of each capture group's **first** match in the string

```
pattern_cg = r"([0-9]{3})-([0-9]{2})-([0-9]{4})"
df["SSN"].str.extract(pattern_cg)
```

	SSN
0	987-65-4321
1	forty
2	123-45-6789 bro or 321-45-6789
3	999-99-9999

	0	1	2
0	987	65	4321
1	NaN	NaN	NaN
2	123	45	6789
3	999	99	9999

`ser.str.extractall(pattern)` docs

Returns a multi-indexed DataFrame of **all** matches for each capture group

```
df["SSN"].str.extractall(pattern_cg)
```

	SSN
0	987-65-4321
1	forty
2	123-45-6789 bro or 321-45-6789
3	999-99-9999

	match	0	1	2
0	0	987	65	4321
2	0	123	45	6789
	1	321	45	6789
3	0	999	99	9999

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Substitution

`re.sub(pattern, repl, text)` docs

Returns text with all instances of `pattern` replaced by `repl`.

```
text = "<div><td valign="top">Moo</td></div>"
pattern = r"<[^>]+>"
re.sub(pattern, '', text) # returns Moo
```

Moo

How it works:

- **`pattern`** matches HTML tags
- Then, sub/replace HTML tags with **`repl=''`** (i.e., empty string)

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Substitution

`re.sub(pattern, repl, text)` docs

Returns text with all instances of `pattern` replaced by `repl`.

```
text = "<div><td valign="top">Moo</td></div>"
pattern = r"<[^>]+>"
re.sub(pattern, '', text) # returns Moo
```

```
Moo
```

How it works:

- **`pattern`** matches HTML tags
- Then, sub/replace HTML tags with **`repl=''`** (i.e., empty string)

`ser.str.replace(pattern, repl, regex=True )` docs

Returns Series with all instances of `pattern` in Series `ser` replaced by `repl`.

```
df["Html"].str.replace(pattern, '')
```

	Html
0	<td valign="top">Moo</td>
1	<a href="http://ds100.org">Link</a>
2	Bold text

```
0           Moo
1          Link
2     Bold text
Name: Html, dtype: object
```


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String function summary

Base Python	re	pandas str
s.lower() s.upper()		ser.str.lower() ser.str.upper()
s.replace(…)	re.sub(…)	ser.str.replace(…)
s.split(…)	re.split(…)	ser.str.split(…)
s[1:4]		ser.str[1:4]
	re.findall(…)	ser.str.findall(…) ser.str.extractall(…) ser.str.extract(…)
'ab' in s	re.search(…)	ser.str.contains(…)
len(s)		ser.str.len()
s.strip()		ser.str.strip()

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Limitations of regular expressions

Writing regular expressions is like writing a program.

- Need to know the syntax well.
- Can be easier to write than to read.
- Can be difficult to debug.

Regular expressions sometimes jokingly referred to as a "**write only language**".

Regular expressions are terrible at certain types of problems:

- For parsing a hierarchical structure, such as JSON, use the `json.load()` parser, not regex!
- Complex features (e.g. valid email address).
- Counting (same number of instances of a and b). (impossible)
- Complex properties (palindromes, balanced parentheses). (impossible)

However, regular expressions are decent at **wrangling text data**.

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More Resources

- https://regexone.com/
  - Hands-on exercises
- https://regex101.com/
  - Checking a regex for a particular problem

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